CHAPTER

2

Soursop Seed: Soursop (*Annona muricata* L.) Seed, Therapeutic, and Possible Food Potential

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List of Abbreviations

AACGs Annonaceous acetogenins
AMS *Annona muricata* seeds
ATP Adenosine triphosphate
ED_{50} Drug dose effective for 50% of exposed population
FA Fatty acid
Hep 2,2,15 Human hepatoma cells infected with hepatitis B virus
Hep G2 Human hepatoma cells
IC50 Drug concentration required for 50% inhibition in vitro
THF Tetrahydrofuran
THP Tetrahydropyran

Introduction

Soursop (*Annona muricata* L) is a plant native to South America that is widely distributed in tropical and subtropical regions of the world; it belongs to the Annonaceae family and its fruits are widely appreciated for their organoleptic characteristics. The family includes several genera characterized by the presence of metabolites with important biological activities. AMS make up approximately 5% of ripe fruit weight. In recent years, scientific

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studies have confirmed the presence of alkaloids, acetogenins, and cyclopeptides, which are compounds of great importance that have attracted pharmacological interest. AMS also have a significant amount of oil whose composition and physical and chemical properties make it potentially attractive in the food sector.

Botanical Descriptions

A. muricata is an annonacea belonging to the Magnoliales Order and the Magnoliopsida Class. It is a tree from 4 to 7 m high, with smooth bark. Its leaves are long and pale green and its flowers have yellow-green fleshy petals. Its fruits are large, oval or heart shaped, with small green thin-shelled spines; their size ranges from 10 to 30 cm long and 15 cm wide, and they can weigh up to 2.5 kg. They have white flesh that is creamy and juicy. The seeds are small and numerous, dark-colored, and of compact consistency. They have a rigid coating that contains a kernel-like almond. At present, there is no information on the number of world varieties. Coarsely, soursop types were classified into sweet, subacid, or sour; however, in just one region of Puerto Rico, 14 varieties have been distinguished, and it is believed that in the other producing regions of the world there are many more.[1]

Historical Cultivation and Usage

Cultivated in Mexico already before the Spaniards arrived, the soursop has been highly appreciated for its edible fruit. It was distributed very early in the warm lowlands of eastern and western Africa, southeastern Asia, and China, where it was and continues to be commonly grown on a small scale or as a backyard tree.[2] It was first used as fresh fruit and later in various food products processed either by hand or on an industrial scale. Historical information on the medicinal use of AMS is scarce, for it was not until the 20th century when its acaricidal property was first reported.

Present-day Cultivation and Usage

The soursop lives in tropical regions with warm weather ranging from sea level to 500 m high. In Mexico, for example, it is cultivated from the state of Sinaloa to the state of Chiapas and from the state of Veracruz to the Yucatan Peninsula along the Gulf of Mexico in small extensions up to 20 *ha*. In many other parts of the world, it is still grown as backyard trees. There are no data available for world production, imports, and exports. In Surinam, this fruit yields 43 kg/tree and 278 trees/*ha*. The commercial plantations are limited to the Philippines, the Caribbean, and South America.

At this time, it is used mainly as fresh and industrialized fruit. The leaves are used commercially for therapeutic uses. Extracts from the seed powder are presently used as an effective insecticide against lice, worms, and aphids, as well as a larvicide.[1,3]

Applications in Health Promotion and Disease Prevention

Annona muricata L. Seed and Its Therapeutic Potential

In traditional medicine it is known that the bark, root, leaf, fruit, and seed of the fruit of *Annona muricata* are used for various medical purposes in the tropics. The crushed seeds are used mainly against internal and external parasites, worms, and lice.[1,3] The studies have been carried out in general with crude extracts of organic solvents such as ethanol and n-hexane, and some of them have allowed to identify their bioactivities as well as to evaluate the synergism of their compounds. For example, ethanolic extracts of AMS have demonstrated their power of inhibition on the growth of larvae of *Spodoptera litura* (Noctuidae),[4] *Aedes aegypti*,[5a–5d] and *Culex quinquefasciatus*, a vector of filariasis a set of infectious diseases that affect the lymphatic system and the skin, of great epidemiological importance in low-income countries.[5d,5e]

Komansilan et al.[5c] showed that the n-hexane fraction was the most effective and toxic against the mosquito *Aedes aegypti*, with a LC_{50} of 73.77 ppm. The extract analyzed by GC-MS showed the relative abundance of methyl palmitate (39.93%), methyl oleate (25.05%), and methyl stearate (25.71%). Ranisaharivony et al.[5d] reported the synergistic activity of AACGs annonacin, murisolin, and annonacinone from AMS extracts and the catalytic hydrogenation of annonacin, which resulted in a more effective mixture of diastereomers than annonacin alone, against the larvae of *C. quinquefasciatus*.

The synergism between the alcoholic extracts of AMS and *Piper nigrum* in larvicidal evaluations realized by Grzybowski et al.[5b] resulted in an efficient, simple, and less toxic formula from AMS. In addition, the antimalarial (*Plasmodium falciparum*) and leishmanicidal activities of the AMS extracts have been evaluated by Boyom et al.[5f] and Vila-Nova et al.,[5g,5h] respectively.

In other tests conducted by Rizki et al.[5i] the ethanolic extract of AMS showed an hypoglycemic effect, finding that with a dose of 400 mg/kg the blood glucose levels were significantly decreased in the test subjects.

On the other hand, the extracts of AMS have also been evaluated in their effectiveness for the control of pests, which indirectly also affect human health, such is the case of the studies that corroborated the effect against *Sitophilus zeamais*, which often infest the stored rice,[5j] and in the cases of cabbage moth *Plutella xylostella*,[5k] and the aphid *Brevicoryne brassicae*, which affects crops of *Brassica oleracea*.[5l]

The Annonaceae family is characterized, in general, as presenting an exclusive group of secondary metabolites of the family of AACGs, compounds which are also found in large numbers in the AMS.[6–8a,8b]

The AACGs are an important group of long-chain FA derivatives, C-30 to C-37, usually with a terminal γ-lactone, saturated or unsaturated. Sometimes the FAs are one to three rings of THF or THP, adjacent or not, which are occasionally replaced with epoxide rings or double bonds near the middle of the aliphatic chain. Accordingly, there can be 7 different types of terminal γ-lactones (L-A to L-F), 10 skeletons with THF and/or THP (T-A to T-H, with 3 subtypes of T-G) and mono-, di-, or triepoxides.[6-8a,8b] Fig. 2.1 shows the general structure of an AACG. The THF ring system can be simple (one ring), adjacent (two rings), or not adjacent, with hydroxyl groups on both sides or not; this creates chiral centers in the molecules, making it very complex to elucidate the structures. Many AACGs isolated as pure compounds are

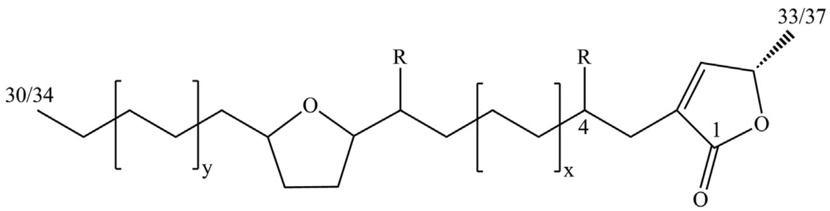

FIGURE 2.1 General structure of an AACG. The AACGs are an important group of long-chain FA derivatives, C-30, C-32, or C-34, usually with a terminal γ-lactone, saturated or unsaturated.

actually a mixture of diastereomers. However, the synthesis of some AACGs makes it possible to assign the chemical configurations of the corresponding natural AACGs unequivocally.[8a–8c]

A little more than 400 compounds have been reported in the literature since the discovery of uvaricin in 1982 all from various genera exclusive to the Annonaceae family,[7] although only some types of structures are present in the AACGs isolated from the AMS.[3] To date, the AACGs that have been isolated from the seeds of soursop include a terminal γ-lactone and THF in their structure; a few have epoxides, yet there are none with adjacent THP or THF.[8a,8b]

More recently, annoreticuin-9-one,[8d] murisolin, *cis*-annoreticulin, and sabadelin, as well as a mixture of β-sitosterol and stigmasterol in a ratio of 1:3, were isolated from AMS.[8e]

Various pharmacological investigations have shown that these metabolites possess antitumoral, antidiarrheal, larvicidal, antimalarial, pesticidal, and fungicidal activities, especially with regard to their antitumoral properties *in vitro*. Some are among the most potent inhibitors of complex I (NADH: ubiquinone oxidoreductase) in the system of mitochondrial electron transport between said complex and the NADH-oxidase in the plasma membrane characteristic of cancer cells; these actions induce apoptosis (programmed cell death), perhaps following the deprivation of ATP. The decrease in ATP is especially toxic to multidrug resistant tumor cells as well as to pesticide-resistant insects that possess ATP-dependent systems of xenobiotic flows, so said metabolites can be regarded as extraordinary antitumor agents and pesticides, especially for inhibiting resistance mechanisms requiring an ATP-dependent flow.[9a]

The AACGs may be among the most potent cytotoxic agents known, for example, trilobacin and asiminocin, AACGs isolated from *Asimina triloba*, have demonstrated values of $DE_{50} < 10^{-12}$ μg/mL in several human tumor cell lines.[9a] A little over 40 AACGs showing cytotoxic activity have been isolated from the AMS, among which *cis*-annonacin may be mentioned as displaying a selective cytotoxicity toward colon adenocarcinoma cells (HT-29), 10,000 times more potent than that of adriamycin ($IC_{50} = 1.0 \times 10^{-8}$ μg/mL, $IC_{50} = 5.1 \times 10^{-4}$ μg/mL, respectively); *cis*-annonacin-10-one showed the same range as adriamycin toward the same type of cell line ($IC_{50} = 9.0 \times 10^{-4}$ μg/mL).[6] The acetogenin annoreticuin-9-one exhibited cytotoxic activity against human pancreatic tumor PACA-2 cell lines, human prostate adenocarcinoma PC-3, and lung carcinoma A-549.[8d] In addition, the anticancer and antitumor activity of AMS AACGs has been shown in cases of oral KB cancer, gallbladder tumor, and toxicity against human hepatoma cells.[9b]

A recent study on the phytoestrogenic, hypocholesterolemic, and antioxidant activities demonstrated by the ethanolic extract of AMS suggests the possible utility of AMS to counteract or eliminate cancerous tumors given the close relationship between the levels of estrogen, cholesterol, and, of course, free radicals, with the incidence of cancer.[9c]

The bioactivity possessed by these compounds has made it possible to obtain a patent for the isolation, identification, and antitumoral use of AACGs from the AMS, those include muricin A, B, C, D, E, F, and G, among which the most effective were muricin D ($IC_{50} = 6.6 \times 10^{-4}$ μg/mL for Hep G_2 and $IC_{50} = 4.8 \times 10^{-2}$ for Hep 2,2,15); muricin F ($IC_{50} = 4.28 \times 10^{-2}$ μg/mL Hep G_2 and $IC_{50} = 3.86 \times 10^{-3}$ μg/mL for Hep 2,2,15); and longifolicin ($IC_{50} = 4.04 \times 10^{-4}$ μg/mL for Hep G_2 and $IC_{50} = 4.9 \times 10^{-3}$ μg/mL for Hep 2,2,15). However, both muricin A and muricin B showed selective activity toward Hep 2,2,15 cell line with $IC_{50} = 5.13 \times 10^{-3}$ μg/mL and $IC_{50} = 4.29 \times 10^{-3}$ μg/mL, respectively.[10] Table 2.1 shows some of the AACGs isolated from the AMS with cytotoxic activity conform to the structure shown in Fig. 2.2; Table 2.2 presents some of the patented AACGs showing substituents according to Fig. 2.3, and Fig. 2.4 shows the muricin I structure that exemplifies AACGs found with double bonds.

TABLE 2.1 Some AACGs With Biological Activity Isolated From AMS.

	x	y	z	R	R_1	*	R_2	References
Cis-Annonacin	2	2	5	OH, 4*R*	OH, 10*R*	19*S*	OH, 20*S*	6
Corossoline	2	2	5	H	OH, 10*R*	19*R*	OH, 20*R*	8b,10
Corossolone	2	2	5	H	=O	19*R*	OH, 20*R*	8b,10
Longifolicin	2	1	6	H	OH, 10*R*	17*R*	OH, 18*R*	8b,10
Murisolin	2	2	5	OH, 4*R*	H	19*S*	OH, 20*S*	8b
Goniothalamicin	2	1	6	OH, 4*R*	OH, 10*R*	17*S*	OH, 18*S*	6

x, y, z: ethylene number in Fig. 2.2; R, R_1, R_2: substituents in Fig. 2.2.

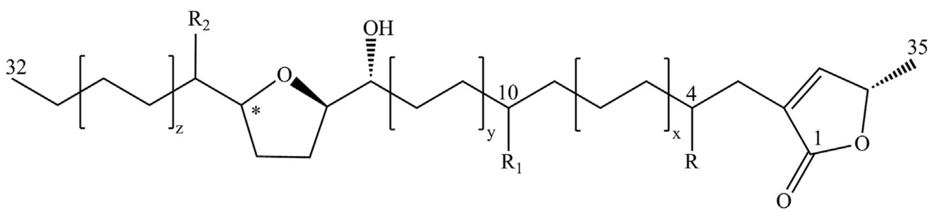

FIGURE 2.2 Basic structure for AACG of Table 2.1. In the figure, when is substituted R, R1, and R2 by H/OH; and *x*, *y*, and z for 2, 2, and 5, respectively, it has the chemical structure of *cis*-annonacin.

TABLE 2.2 Some Patented AACGs With Biological Activity Isolated From AMS (10).

	x	y	z	R	R_1	R_2	References
Muricatetrocin	6	1	9	OH, 19*R*	OH, 20*R*	H	8b,10
Muricin A	9	3	2	OH	OH	H	8b,10
Muricin C	11	1	4	OH	OH	H	8b,10
Muricin D	9	1	4	OH	OH	H	8b,10
Muricin E	6	2	5	H	OH	OH	8b,10

x, y, z: methylene and ethylene number in Fig. 2.3; R, R_1, R_2: substituents in Fig. 2.3.

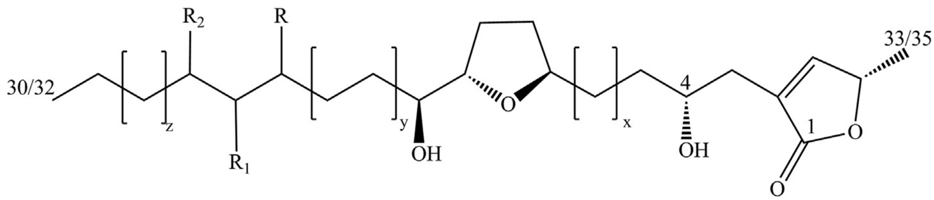

FIGURE 2.3 Basic structure for AACG of Table 2.2. In the figure, when is substituted R, R_1, and R_2 by OH, OH, and H; and *x*, *y*, and z for 6, 1, and 9, respectively, it has the muricatetrocin A chemical structure.

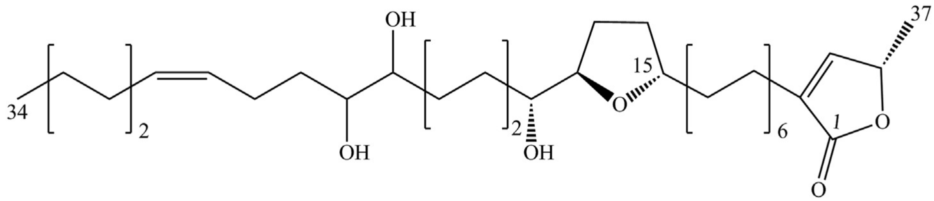

FIGURE 2.4 Muricin I structure exemplifies AACGs found with double bonds.

In addition, the cyclopeptides annomuricatin A, annomuricatin B, and annomuricatin C have been isolated from AMS;[11a] however, X-ray diffraction studies showed that both annomuricatin A and annomuricatin C are the same compound.[11b] These types of compounds have demonstrated activity in other species of *Annona*[11a] and so has lectin that was found to possess inhibitory activity against the fungal pathogens *Fusarium oxysporum*, *F. solana*, and *Colletotrichum musae*, which can produce Fusarium wilt (Panama disease), keratomycosis, and anthracnose, in addition to red cell agglutinating activity.[12]

Accordingly, the AMS is a source of bioactive compounds (AACGs) with immense possibilities for pharmaceutical uses.

Annona muricata Seed and Its Possible Potential for Food Use

The use of soursop fruit in the production of foodstuffs results in various so-called waste materials, one of which is the seeds. Studies of the AMS in its proximal composition have shown that it contains proteins, carbohydrates, ash, and oil in significant amounts. The reports are scarce and contain wide ranges of values that reflect the need for further studies regarding varieties and their degree of variability. Table 2.3 shows the composition and property ranges of AMS and its oil as reported in pertinent literature.

There are few studies that have directly explored AMS possibilities as raw material in the food sector, mainly because there is knowledge of the existence of toxic compounds in it. Awan et al.,[13] and particularly Fasakin et al.,[14] analyzed the AMS with the purpose of highlighting the nutritional potential of the seed; other authors have studied it for the chemical and physical characteristics and the thermal and phase behavior of its oil,[15–17a] with the objective of contributing to knowledge about the reuse of soursop wastes or of characterizing some of its most important components.

TABLE 2.3 Reported Composition and Property Ranges of AMS and Oil

Properties	Value Range	Units	References
Seeds in fruit	4.0–5.4	%, as-is basis	14,17a
AMS composition			
Moisture	2.17–34.6		14,17a,[a]
Protein	2.4–27.3	% of kernel, db	14,15
Crude fiber	5.2–43.4		14,15,[b]
Ash	1.3–13.6		14,15,17c
Raw oil	18.3–37.7		13,17a,17c
Carbohydrates	11.5–34.1		17c,[b]
Polyphenols	3.1	mg GAE/g	17c
AMSO composition			
Myristic (C14:0)	0.06	% of total FA	[c]
Pentadecylic (C15:0)	0.4		[d]
Palmitic (C16:0)	19–25.5		16,17a
Palmitoleic (C16:1)	1.34–17.29		16,17a,[d,e]
Margaric (C17:0)	0.07–0.11		[c,d]
Stearic (C18:0)	3.3–7.90		16,17a,[c,e]
Oleic (C18:1)	39.18–44.0		16,17a,[c]
Linoleic (C18:2)	21.74–35.9		16,17a[d,e]
Linolenic (C18:3)	1.16		[c]
Arachidic (C20:0)	0.42–1.19		[c,e]
Gadoleic (C20:1)	0.07–0.16		17c,[c]
Gandoic (C20:1)	0.16		[c]
Behenic (C22:0)	0.08–0.10		17c,[c]
Lignoceric (C24:0)	0.10–0.15		17c,[c]
SFA	24–31.5		16,17a
UFA	68.5–76.82		16,17a,[c]
α-Tocopherol	12.5	mg/kg oil	[c]
AMSO properties			
Specific gravity (20°C)	0.9281		[c]
Refraction index	1.451–1.468		15,17a,[c]
Saponification index	100–227		13,15

(*Continued*)

TABLE 2.3 Reported Composition and Property Ranges of AMS and Oil—cont'd

Properties	Value Range	Units	References
Peroxide value	8.5	mEq O_2/kg oil	[c]
Iodine value	87–111		13
Acid value	3.13		[c]
SFC at 10°C	1.3	%	17a

[a]*Nzekwe ABC, Nzekwe FN. Proximate analysis & characterization of the seed and oil of* Annona muricata *(soursop).* The Nigerian Journal of Research and Production. *2011;18:1–3.*
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[c]*Elagbar ZA, Naik RR, Shakya AK, Bardaweel SK. Fatty acids analysis, antioxidant and biological activity of fixed oil of* Annona muricata *L. seeds.* Journal of Chemistry. *2016; 1–6. https://doi.org/10.1155/2016/6948098.*
[d]*Navaratne SB, Subasinghe JS. Determination of fatty acid profile and physicochemical properties of watermelon and soursop seed oils.* European International Journal of Applied Science and Technology. *2014; 1:26–32.*
[e]*Nwaehujor IU, Olatunji GA, Afolayan SS, et al. Physicochemical properties of the seed oil of* Annona muricata *grown in Ilorin, Kwara State.* Equijost. *2016; 4:1–4.*

The chemical composition of the kernel seed, according to research conducted to date, may be interesting from the standpoint of food. For example, substances that are important for animal feed, such as protein and fiber, minerals and oil, can be found in the seed at levels from 2.4 to 27.3%, 5.2 to 43.4%, 1.3 to 13.6%, and 18.3 to 37.7%, respectively.

In recent times, there has been extensive research to locate and expand nonconventional sources of fats and oils for the purpose of finding applications for them in various fields. The AMS has an important amount of oil. Composed of oleic (39.18–53.92%), linoleic (21.74–35.90%), palmitic (16.39–25.50%), and stearic (3.30–7.90%) as its main FAs, it looks like the oils from conventional oilseeds. The reported values for its refraction index, saponification index, and iodine value range from 1.451 to 1.468, 100 to 227, and 87 to 111, respectively, and studies of its phase and thermal behavior have showed that it has the characteristics of common table or cooking oils; the yellowish oil remains liquid at warm ambient and refrigeration temperatures, its solid fat contents at 10°C being around 1% (see Table 2.3 for references).

Even with all these considerations, it is clear that any application of the AMS or its derived components must take into account the characteristic toxicity of raw AMS. Notwithstanding the bromatological characteristics of the seed and its significant oil yield and physicochemical properties, its use for human consumption should be viewed with caution. The usability of AMS and its oil undoubtedly require increased knowledge, particularly in regard to food applications. The reviewed bioactive compounds present in the seed are of toxic and pharmacological importance; consequently, they limit the current use of raw AMS in foods. The processing of AMS to isolate and obtain components of interest to the food industry, such as proteins, oil, or otherwise, also requires more and deeper studies aimed at extraction, purification, toxicological and nutritional evaluations, etc.

In this regard, recently AMSO has been used experimentally in the treatment of skin wounds, with interesting results in its positive effect on the healing process.[17b]

Pinto et al.[17c] studied a method to extract AMSO and reduce its toxicity, using a mixture of chloroform–methanol, of which they were able to obtain a solid fraction of a toxic precipitate, presumably with the majority of AACGs, and a supernatant corresponding to the oil.

The AMSO thus separated from that solid phase and obtained through this method showed no toxicity, but a hepatoprotective effect, through *in vivo* evaluations with mice. It had an FA profile similar to that reported by other authors in AMS extractions with other organic solvents, maintaining the perspective of its possible use as food with the health benefits because of its high content of monounsaturated and polyunsaturated FA.

On the other hand, the possible presence of active compounds such as the AACGs and others from AMS in the crude oil opens the possibility too, to be used in insecticide and therapeutic applications, such as diabetes type 1, among others, in accordance with the proven activity of these compounds.[5l,17c,17d]

Adverse Effects and Reactions, Allergies, and Toxicity

Soursop seed is not edible and traditionally has been known to be toxic because of the presence of the described compounds; therefore, its components and derivatives destined for human consumption must be subject to careful research protocols to ensure their toxicological safety. Even when it is used as a drug, in its whole form or from crude extracts, it must be handled with care because some of its bioactive components have not been fully evaluated for their allergenic or toxicological cumulative effect. One example is annonacin, which has been isolated from most parts of the plant, including seeds and fruit pulp. It has been shown by clinical studies with rats and epidemiological studies in human populations (like those from the Caribbean island of Guadeloupe and those from New Caledonia) with high and prolonged intakes of soursop fruit pulp that annonacin is associated with the occurrence of a neurodegenerative disease known as atypical parkinsonism. This finding, together with the known effects of the AACG on mitochondrial toxicity, makes the consumption of the AMS, its concentrates, and derivates without purification a risky matter.[18–20] The latter, notwithstanding the research efforts to evaluate the safety of some fractions of AMS for food purposes, as in the case of AMSO, it is considered that the results to date are not conclusive, yet.

Summary Points

- Soursop, *Annona muricata* L., is a plant native to South America and is widely cultivated throughout the tropical regions of the world.
- *A. muricata* L. has shown pharmacological activity in its different parts.
- *Annona muricata* seeds make up about 5% of the fruit and are now considered an industrial processing waste.
- Annonaceous acetogenins are the most studied bioactive metabolites from *Annona muricata* seeds and they have high cytotoxic activity with potential therapeutic applications.
- Other important components of the *Annona muricata* seeds have also been studied from a food perspective.
- *Annona muricata* seeds have a significant content of oil which has a composition and properties similar to those oils of conventional sources.
- Any component derived from *Annona muricata* seeds to alimentary consumption must be previously studied about their possible toxicological risk.

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